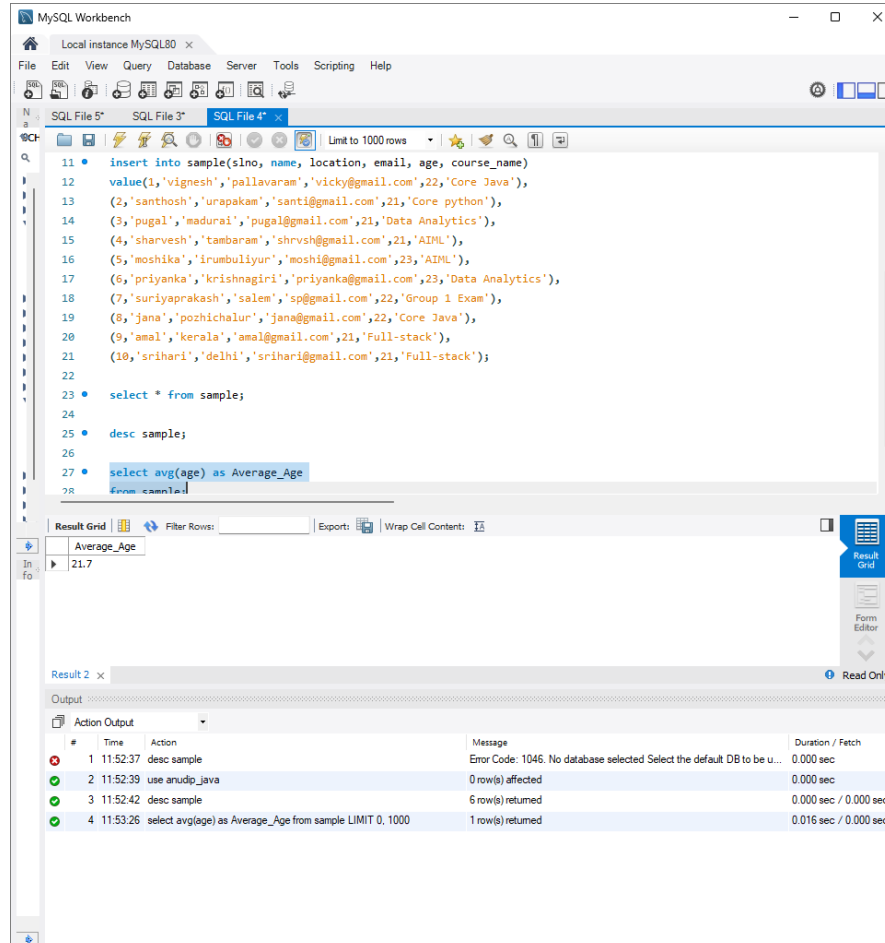


Task

Description

#Regarding the Last Task (Database Creation and Access), use aggregate function to show the average of ages



The screenshot shows the MySQL Workbench interface. The SQL Editor contains the following queries:

```
11 • insert into sample(slno, name, location, email, age, course_name)
12 value(1,'vignesh','pallavaram','vicky@gmail.com',22,'Core Java'),
13 (2,'santhosh','urapakam','santi@gmail.com',21,'Core python'),
14 (3,'pugal','madurai','pugal@gmail.com',21,'Data Analytics'),
15 (4,'sharvesh','tambaram','shrvsh@gmail.com',21,'AIML'),
16 (5,'moshika','irumbuliyur','moshi@gmail.com',23,'AIML'),
17 (6,'priyanka','krishnagiri','priyanka@gmail.com',23,'Data Analytics'),
18 (7,'suriyaprakash','salem','sp@gmail.com',22,'Group 1 Exam'),
19 (8,'jana','pozhichalur','jana@gmail.com',22,'Core Java'),
20 (9,'amal','kerala','amal@gmail.com',21,'Full-stack'),
21 (10,'srihari','delhi','srihari@gmail.com',21,'Full-stack');
22
23 • select * from sample;
24
25 • desc sample;
26
27 • select avg(age) as Average_Age
from sample;
```

The Result Grid shows the output of the last query:

Average_Age
21.7

The Output tab shows the execution log:

#	Time	Action	Message	Duration / Fetch
1	11:52:37	desc sample	Error Code: 1046. No database selected Select the default DB to be u...	0.000 sec
2	11:52:39	use anudp_java	0 row(s) affected	0.000 sec
3	11:52:42	desc sample	6 row(s) returned	0.000 sec / 0.000 sec
4	11:53:26	select avg(age) as Average_Age from sample LIMIT 0, 1000	1 row(s) returned	0.016 sec / 0.000 sec

add the column for the particular table (cgpa)

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following queries:

```
16 (5,'moshika','irumbuliyur','moshi@gmail.com',23,'AIML'),
17 (6,'priyanka','krishnagiri','priyanka@gmail.com',23,'Data Analytics'),
18 (7,'suriyaprakash','saalem','sp@gmail.com',22,'Group 1 Exam'),
19 (8,'jana','pozhichalur','jana@gmail.com',22,'Core Java'),
20 (9,'amal','kerala','amal@gmail.com',21,'Full-stack'),
21 (10,'srihari','delhi','srihari@gmail.com',21,'Full-stack');
22
23 • select * from sample;
24
25 • desc sample;
26
27 • select avg(age) as Average_Age
28   from sample;
29
30 • alter table sample
31   add cgpa float(4,2);
32
```

The Result Grid shows the structure of the 'sample' table:

Field	Type	Null	Key	Default	Extra
sno	int	NO	PRI		
name	varchar(20)	YES			
location	varchar(20)	YES			
email	varchar(20)	YES	UNI		
age	varchar(20)	YES			
course_name	varchar(20)	YES			
cgpa	float(4,2)	YES			

The Action Output shows the execution of the queries:

#	Time	Action	Message	Duration / Fetch
1	11:52:37	desc sample	Error Code: 1045: No database selected Select the default DB to be u...	0.000 sec
2	11:52:39	use anudip_java	0 row(s) affected	0.000 sec
3	11:52:42	desc sample	6 row(s) returned	0.000 sec / 0.000 sec
4	11:53:26	select avg(age) as Average_Age from sample LIMIT 0, 1000	1 row(s) returned	0.016 sec / 0.000 sec
5	11:55:24	desc sample	6 row(s) returned	0.000 sec / 0.000 sec
6	11:56:33	alter table sample add cgpa float(4,2)	0 row(s) affected, 1 warning(s): 1681 Specifying number of digits for flo...	0.032 sec
7	11:57:09	alter table sample add cgpa decimal(4,2)	Error Code: 1060: Duplicate column name 'cgpa'	0.000 sec
8	11:57:16	desc sample	7 row(s) returned	0.016 sec / 0.000 sec

maximum mark?

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following queries:

```
23 • select * from sample;
24
25 • desc sample;
26
27 • select avg(age) as Average_Age
28   from sample;
29
30 • alter table sample
31   modify cgpa float(3,2);
32
33 • update sample
34   set cgpa = 7.92
35   where sno in (4) ;
36
37 • select max(cgpa) as Maximum_Mark
38   from sample;
39
```

The Result Grid shows the result of the last query:

Maximum_Mark
7.92

The Action Output shows the execution of the queries:

#	Time	Action	Message	Duration / Fetch
10	11:59:20	select * from sample LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec
11	12:00:45	alter table sample modify cgpa float(3,2)	0 row(s) affected, 1 warning(s): 1681 Specifying number of digits for fl...	0.016 sec
12	12:00:53	alter table sample set cgpa = 9.02 where sno = 1	Error Code: 1064: You have an error in your SQL syntax; check the ...	0.000 sec
13	12:02:10	update sample set cgpa = 9.02 where sno = 1	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec
14	12:03:03	update sample set cgpa = 8.72 where sno in (2,5,7,8,10)	5 row(s) affected Rows matched: 5 Changed: 5 Warnings: 0	0.000 sec
15	12:03:39	update sample set cgpa = 8.92 where sno in (3,5,9)	3 row(s) affected Rows matched: 3 Changed: 3 Warnings: 0	0.000 sec
16	12:04:21	select * from sample LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec
17	12:05:04	update sample set cgpa = 7.92 where sno in (4)	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.015 sec
18	12:05:38	select max(cgpa) as Maximum_Mark from sample LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec

minimum mark

The screenshot shows the MySQL Workbench interface with a SQL editor containing the following queries:

```
26
27 select avg(age) as Average_Age
28 from sample;
29
30 alter table sample
31 modify cgpa float(3,2);
32
33 update sample
34 set cgpa = 7.92
35 where slno in (4) ;
36
37 select max(cgpa) as Maximum_Mark
38 from sample;
39
40 select min(cgpa) as Minimum_Mark
41 from sample;
42
```

The Results pane shows the output of the last query, displaying a single row with the value 7.92 for the Minimum_Mark.

The Output pane shows the execution log with the following entries:

#	Time	Action	Message	Duration / Fetch
11	12:00:45	alter table sample modify cgpa float(3,2)	0 row(s) affected. 1 warning(s): 1681 Specifying number of digits for fl...	0.015 sec
12	12:00:53	alter table sample set cgpa = 9.02 where slno = 1	Error Code: 1064. You have an error in your SQL syntax; check the ..	0.000 sec
13	12:02:10	update sample set cgpa = 9.02 where slno = 1	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec
14	12:03:03	update sample set cgpa = 8.72 where slno in (2,5,7,8,10)	5 row(s) affected Rows matched: 5 Changed: 5 Warnings: 0	0.000 sec
15	12:03:39	update sample set cgpa = 8.92 where slno in (3,6,9)	3 row(s) affected Rows matched: 3 Changed: 3 Warnings: 0	0.000 sec
16	12:04:21	select * from sample LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec
17	12:05:04	update sample set cgpa = 7.92 where slno in (4)	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.015 sec
18	12:05:38	select max(cgpa) as Maximum_Mark from sample LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
19	12:06:33	select min(cgpa) as Minimum_Mark from sample LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec

second highest Mark ?

The screenshot shows the MySQL Workbench interface with a SQL editor containing the following queries:

```
32
33 update sample
34 set cgpa = 7.92
35 where slno in (4) ;
36
37 select max(cgpa) as Maximum_Mark
38 from sample;
39
40 select min(cgpa) as Minimum_Mark
41 from sample;
42
43 select max(cgpa) as second_max_mark from sample
44 where cgpa < (select max(cgpa) from sample);
45
```

The Results pane shows the output of the last query, displaying a single row with the value 8.92 for the second_max_mark.

The Output pane shows the execution log with the following entries:

#	Time	Action	Message	Duration / Fetch
14	12:03:03	update sample set cgpa = 8.72 where slno in (2,5,7,8,10)	5 row(s) affected Rows matched: 5 Changed: 5 Warnings: 0	0.000 sec
15	12:03:39	update sample set cgpa = 8.92 where slno in (3,6,9)	3 row(s) affected Rows matched: 3 Changed: 3 Warnings: 0	0.000 sec
16	12:04:21	select * from sample LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec
17	12:05:04	update sample set cgpa = 7.92 where slno in (4)	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.015 sec
18	12:05:38	select max(cgpa) as Maximum_Mark from sample LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
19	12:06:33	select min(cgpa) as Minimum_Mark from sample LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
20	12:09:08	select * from sample LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec
21	12:09:33	select max(cgpa) from sample where cgpa < (select max(cgpa) from s...	1 row(s) returned	0.000 sec / 0.000 sec
22	12:10:05	select max(cgpa) as second_max_mark from sample where cgpa < (s...	1 row(s) returned	0.000 sec / 0.000 sec